Additional file 1. Supplementary material for "Drift Detection, Diagnosis and Reliability Analysis for Deployed Machine Learning Models"

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Section, figure, table and equation numbers without the prefix S refer to the main paper.

Five topics that the main paper only touches on are expanded here. Section S1 asks how many delayed labels the audit needs before its diagnosis settles. Section S2 breaks the remediation triage down by real setting, while Section S3 records how the relationship codes of the American Community Survey (ACS) were matched across their 2019 redesign. Section S4 presents the dashboard in which the pipeline displays its results. The last section, Section S5, shows how the benchmark results vary over seeds and states.

S1 Size of the delayed audit

The audit stage of Section 2.1.3 needs late labels. Labels cost money, so it matters how many are enough. The paper uses 600 labels throughout. Fig. S1 shows how the share of correctly diagnosed windows changes with the size of the audit on ACS Income design A with logistic regression (LR).

With 200 labels, concept drift is identified in 80% of windows and label noise in 79%. With 600 labels the two rates reach 95% and 87%. Going on to 1000 labels adds little (96% and 82%). The 600 labels of the paper therefore sit where more labels stop paying off. The control windows are diagnosed correctly in the same share of cases at every audit size.

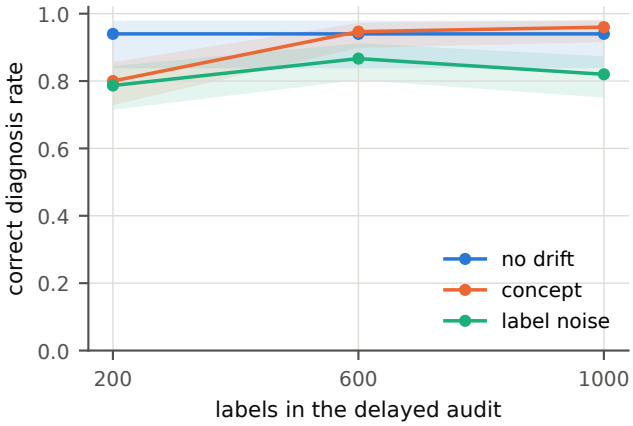


Fig. S1 Share of correctly diagnosed windows on ACS Income design A with LR against the number of audit labels. The shaded bands are 95% Wilson intervals

S2 Remediation triage by setting

Section 3.5 of the paper summarizes how often the triage of Section 2.1.5 gives the right advice. Table S1 breaks this summary down by real setting and model family, so that the source of each right or wrong verdict can be seen.

Where an earlier labeled window exists, retraining on it helps in almost every temporal window. The triage escalates in those windows. The geographic windows of the census tasks have no earlier window. There, importance weighting rarely helps and the triage mostly advises against a retrain. Most of the remaining wrong verdicts come from CLEAR-10 with LR and from the geographic windows of ACS Employment. The simple rule in the last column is right about as often as the triage, which is why the paper reports the triage as a negative result.

S3 Relationship codes across the 2019 redesign

The ACS windows run from 2014 to 2023, but one of their features changed its coding halfway through. From 2019 the Census Bureau replaced the relationship variable RELP with RELSHIPP, whose codes split spouses and partners by sex (U.S. Census Bureau 2020). To compare windows before and after 2019, we map the RELSHIPP codes back to RELP by matching the code labels of the 2018 and 2019 data dictionaries. Table S2 lists the mapping as the code applies it.

RELP 11 (roomer or boarder) has no successor, so it never occurs from 2019 on. The loss of this category is itself a documented change that the localization can pick up.

Table S1 Remediation triage on the real windows. n is the number of windows; Significant gains counts the windows in which a strategy significantly raised holdout accuracy, by importance weighting (IW), retraining on the most recent labeled window (Recent) or retraining on labels of the window itself (Labels); n/a marks settings without an earlier window

			Significant gains			Triage			Rule
Model	Settings	n	IW	Recent	Labels	Escalated	Truly helps	Correct	Correct
LR	ACS Income, temporal	8	4/8	7/7	5/8	7	7	8/8	8/8
	ACS Income, states	4	0/4	n/a	4/4	0	0	4/4	4/4
	ACS Employment, temporal	8	4/8	7/7	2/8	7	7	8/8	8/8
	ACS Employment, states	4	2/4	n/a	1/4	1	2	3/4	2/4
	CLEAR-10	9	0/9	4/8	6/9	4	4	7/9	5/9
	Gas sensor	8	1/8	7/7	8/8	7	7	8/8	8/8
GB	ACS Income, temporal	8	0/8	7/7	4/8	7	7	8/8	8/8
	ACS Income, states	4	0/4	n/a	3/4	0	0	4/4	4/4
	ACS Employment, temporal	8	0/8	7/7	0/8	6	7	7/8	8/8
	ACS Employment, states	4	0/4	n/a	0/4	0	0	4/4	4/4
	CLEAR-10	9	0/9	8/8	9/9	8	8	9/9	9/9
	Gas sensor	8	1/8	5/7	8/8	7	6	7/8	7/8

Escalated counts the windows in which the triage recommended a retrain, Truly helps those in which a cheap retrain did help and Correct the windows in which the advice was right. Rule counts the windows in which the simple rule of retraining exactly when an earlier labeled window exists was right

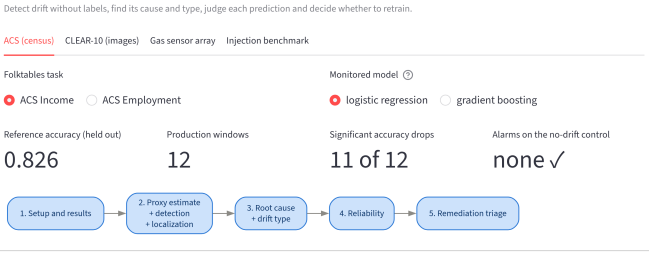
Table S2 Mapping from RELSHIPP (2019 on) back to RELP (2014 to 2018)

RELP	Label (2018)	RELSHIPP codes (2019 on)
0	Reference person	20
1	Husband/wife	21, 23
2	Biological son or daughter	25
3	Adopted son or daughter	26
4	Stepson or stepdaughter	27
5	Brother or sister	28
6	Father or mother	29
7	Grandchild	30
8	Parent-in-law	31
9	Son-in-law or daughter-in-law	32
10	Other relative	33
11	Roomer or boarder	none
12	Housemate or roommate	34
13	Unmarried partner	22, 24
14	Foster child	35
15	Other nonrelative	36
16	Institutionalized group quarters population	37
17	Noninstitutionalized group quarters population	38

S4 The monitoring dashboard

A diagnosis is only useful if the team can read it. Every stage of the pipeline therefore writes its results to one file, which an interactive dashboard presents. Fig. S2 shows two of its views for ACS Income with LR.

(a)



(b) 3b. Drift type

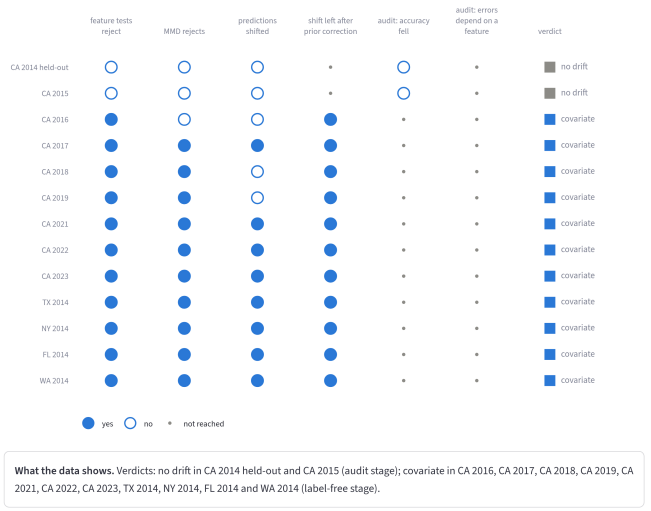


Fig. S2 The dashboard of the pipeline for ACS Income with LR. (a) The overview: dataset, task and model selectors, the headline results and the five steps of the pipeline, each of which opens a detailed view. (b) The drift type step: for every window, the answer to each question of the decision path in Fig. 2 of the paper (filled circle yes, open circle no, small dot not reached) and the resulting verdict, followed by a summary sentence computed from the results

Panel (b) applies the decision path of Fig. 2 to the real California (CA) windows. CA 2015 passes every test of the label-free stage. The KS detector in Table 7 of the paper does raise an alarm on CA 2015, because it corrects only over the numeric features, while the label-free stage tests all features together with the other two blocks in one weighted Holm family (Section 2.1.3). The audit then finds no significant drop

in accuracy, so the verdict is no drift. In CA 2016, CA 2018 and CA 2019 the predicted labels stay where they were. The feature tests still reject and the shift survives the prior correction, so these windows are covariate shift. All other windows answer yes to every label-free question and are covariate shift as well. Each view of the dashboard opens with the question it answers and closes with a summary sentence of this kind, so a team can follow the diagnosis without reading the code.

S5 Spread over seeds and states

The shares in Tables 3 and 4 of the paper pool the windows of all five states and ten seeds. Table S3 shows how much the share of correctly typed windows varies and Table S4 does the same for the design B mechanisms. Each cell gives the mean and standard deviation of the share over the ten seeds, where each seed pools its five states. The interval in brackets treats the state as the cluster: it is a 95% t interval with four degrees of freedom over the five state shares, cut at 0 and 100%. Five clusters make these intervals wide.

References

U.S. Census Bureau (2020) American community survey 2019 PUMS data dictionary. URL https://www2.census.gov/programs-surveys/acs/tech_docs/pums/data_dict/PUMS_Data_Dictionary_2019.csv, accessed 21 October 2026

Table S3 Share of benchmark windows with the correct drift type over seeds and states (%)

Model	Method	Inc. A	Inc. B	Emp. A	Emp. B
LR	Ours (hierarchical tests)	94.3 ± 7.6 [88, 100]	62.4 ± 3.5 [59, 66]	85.7 ± 5.6 [79, 92]	80.5 ± 4.9 [77, 84]
	Input monitoring	30.3 ± 1.0 [30, 31]	20.0 ± 0.0 [20, 20]	30.6 ± 0.5 [30, 31]	25.0 ± 0.0 [25, 25]
	BBSE first	30.5 ± 1.0 [30, 31]	27.7 ± 5.4 [24, 31]	30.5 ± 1.0 [30, 31]	39.2 ± 9.7 [37, 42]
	Error-rate monitoring	52.3 ± 3.8 [50, 55]	48.5 ± 6.8 [45, 53]	52.2 ± 3.8 [50, 54]	62.8 ± 10.7 [59, 66]
	Learned meta-classifier	97.4 ± 1.8 [96, 99]	31.7 ± 3.1 [24, 39]	95.7 ± 1.7 [94, 97]	48.5 ± 7.1 [43, 54]
	Ours without MMD block	95.2 ± 4.9 [90, 100]	62.9 ± 2.5 [59, 67]	86.6 ± 4.0 [81, 92]	81.3 ± 3.4 [77, 85]
	Ours, audit null without confidence	79.1 ± 5.2 [75, 83]	62.9 ± 3.9 [58, 68]	83.7 ± 5.4 [77, 90]	80.7 ± 4.5 [77, 84]
	Ours, audit null with class cells	93.1 ± 8.0 [87, 99]	65.6 ± 4.2 [61, 70]	86.2 ± 6.7 [79, 93]	91.8 ± 5.4 [89, 94]
GB	Ours (hierarchical tests)	94.2 ± 8.0 [88, 100]	63.2 ± 3.6 [60, 66]	90.5 ± 3.5 [84, 97]	81.5 ± 3.5 [78, 85]
	Input monitoring	30.3 ± 1.0 [30, 31]	20.0 ± 0.0 [20, 20]	30.6 ± 0.5 [30, 31]	25.0 ± 0.0 [25, 25]
	BBSE first	30.5 ± 1.0 [30, 31]	28.1 ± 5.9 [25, 31]	30.6 ± 0.5 [30, 31]	40.3 ± 9.2 [38, 43]
	Error-rate monitoring	52.5 ± 3.9 [50, 55]	49.6 ± 7.7 [46, 53]	53.1 ± 2.0 [51, 55]	64.8 ± 9.6 [62, 68]
	Learned meta-classifier	98.3 ± 1.7 [97, 100]	39.2 ± 2.7 [37, 41]	97.5 ± 2.4 [97, 98]	50.7 ± 6.6 [46, 55]
	Ours without MMD block	95.1 ± 5.2 [89, 100]	63.9 ± 2.2 [60, 68]	90.5 ± 3.1 [84, 97]	81.3 ± 4.4 [79, 83]
	Ours, audit null without confidence	78.0 ± 6.2 [75, 81]	62.9 ± 3.3 [60, 66]	85.4 ± 4.3 [80, 91]	82.0 ± 3.2 [79, 85]
	Ours, audit null with class cells	92.9 ± 8.2 [86, 100]	66.5 ± 4.2 [62, 71]	90.6 ± 3.9 [84, 97]	93.7 ± 4.4 [91, 96]

Inc. and Emp. stand for ACS Income and ACS Employment, A and B for the two designs

Table S4 Correct diagnosis by design B mechanism over seeds and states (%)

	ACS Income		ACS Employment	
Mechanism	LR	GB	LR	GB
Covariate: selection bias	99.3 ± 2.1 [97, 100]	99.3 ± 2.1 [97, 100]	100.0 ± 0.0 [100, 100]	100.0 ± 0.0 [100, 100]
Prior: decrease	92.0 ± 9.3 [82, 100]	92.7 ± 8.0 [84, 100]	96.7 ± 6.5 [91, 100]	96.7 ± 6.5 [91, 100]
Concept: oblique region	95.3 ± 12.6 [89, 100]	95.3 ± 12.6 [89, 100]	90.7 ± 13.0 [81, 100]	96.7 ± 6.5 [92, 100]
Concept: income threshold	4.0 ± 4.7 [0, 9]	6.0 ± 5.8 [0, 12]	–	–
Noise: class-conditional	21.3 ± 8.2 [13, 30]	22.7 ± 7.2 [14, 32]	34.7 ± 11.7 [24, 45]	32.7 ± 9.1 [25, 41]